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CLOUD-NATIVE SMART WATER DISTRIBUTION MONITORING AND LEAKAGE DETECTION PLATFORM

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for the Course
of*

Capstone Project

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (BTECH AIDS)**

Submitted by

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Under the Supervision of

Boomija M D

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "CLOUD-NATIVE SMART WATER DISTRIBUTION MONITORING AND LEAKAGE DETECTION PLATFORM" has been carried out by Nitheesh S(Reg. No. 192524329) under the supervision of [Boomija M D] and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Data Science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, Nitheesh S of the Department of Artificial Intelligence and Data Science (AIDS), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "CLOUD-NATIVE SMART WATER DISTRIBUTION MONITORING AND LEAKAGE DETECTION PLATFORM" is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:

Signature of the Student: Nitheesh S (192524329)



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Individual Contribution Statement

(Mandatory for all team projects; recorded here for the individually submitted portion of the work.)

Student Name / Reg No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Nitheesh S (192524329)	System design, AI-based leak detection module, cloud architecture, implementation, testing and report preparation	Designed the cloud-native architecture, edge-to-cloud data pipeline and the AI anomaly-detection module	Carried out functional testing, validation of anomaly-detection accuracy and performance evaluation	Authored all chapters of the capstone project report	100%

ABSTRACT

Urban water utilities lose an estimated 20-30% of treated water to physical leaks and unmetered flow before it reaches consumers, a problem that legacy SCADA systems and manual inspection routines cannot address at the speed or scale that modern distribution networks demand. This capstone project addresses the engineering problem of detecting, localising and responding to leaks and non-revenue water losses in a large, sprawling pipe network in near real time, while also providing utilities with auditable compliance reporting. The proposed solution is a Cloud-Native Smart Water Distribution Monitoring and Leakage Detection Platform that unifies IoT telemetry, GIS pipeline mapping, AI-based anomaly detection, SCADA control and compliance reporting into a single elastically scalable system. The platform follows a layered architecture consisting of a field layer of IoT flow and pressure sensors and SCADA RTUs, an edge and ingestion layer using MQTT/Modbus protocol adapters and a streaming telemetry bus, a cloud platform layer built on Kubernetes microservices with a time-series data lake and an AI/ML anomaly engine, and an application layer offering GIS dashboards, a SCADA control console, and alerting, dispatch and compliance-reporting tools. The methodology combines real-time telemetry ingestion and GIS-based visualisation, statistical and machine-learning anomaly detection for leak localisation and root-cause diagnosis, District Metered Area (DMA) based consumption balancing to expose non-revenue water and theft, and automated incident dispatch with SLA and KPI tracking. The system targets a leak-detection latency of under five minutes and continuous 24/7 autonomous SCADA monitoring and control. The major results establish that the layered cloud-native design successfully consolidates previously siloed monitoring, detection, control and reporting workflows into a single operational view, that anomaly-detection models can flag abnormal flow and pressure patterns with an associated confidence score to prioritise field verification, and that DMA-level consumption comparison provides a practical mechanism for surfacing non-revenue water. The principal conclusion is that integrating telemetry, GIS, AI analytics, SCADA control and compliance reporting into one modular, cloud-native platform closes the gap between detection and action, directly reducing non-revenue water and operating cost while improving response times and regulatory confidence, and that the design is scalable from a single utility zone to an entire city network.

Keywords: Cloud-Native Architecture, Internet of Things (IoT), SCADA, Leak Detection, Anomaly Detection, Non-Revenue Water

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
IoT	Internet of Things	-
GIS	Geographic Information System	-
SCADA	Supervisory Control and Data Acquisition	-
DMA	District Metered Area	-
AI/ML	Artificial Intelligence / Machine Learning	-
MQTT	Message Queuing Telemetry Transport	-
RTU	Remote Terminal Unit	-
NRW	Non-Revenue Water	-
API	Application Programming Interface	-
SLA	Service Level Agreement	-
KPI	Key Performance Indicator	-

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Urban water distribution networks are large, geographically distributed infrastructure systems consisting of pipelines, valves, pumps, storage reservoirs and metering points that must maintain adequate pressure and flow while minimising loss. Traditional utilities rely on periodic manual inspection and legacy Supervisory Control and Data Acquisition (SCADA) systems that provide limited, siloed visibility into network conditions. As cities grow and pipe networks expand, the volume of telemetry data, the number of District Metered Areas (DMAs) and the complexity of coordinating field response increase substantially, creating a strong technical case for a cloud-native, data-driven monitoring and control platform that can unify Internet of Things (IoT) telemetry, Geographic Information System (GIS) mapping, Artificial Intelligence (AI) analytics, SCADA actuation and compliance reporting.

1.2 Need for the Project

The problem needs to be addressed because urban water utilities lose an estimated 20-30% of treated water to physical leaks and unmetered flow before it reaches consumers, representing a significant economic, environmental and operational burden. This loss affects utility operators, municipal authorities, regulators and, ultimately, consumers who bear the cost of non-revenue water (NRW) through tariffs and reduced service reliability. Existing limitations include the inability of legacy SCADA and manual inspection processes to detect leaks early enough, to react to demand shifts in real time, or to scale cost-effectively across sprawling pipe networks. From an engineering significance perspective, closing the gap between leak occurrence and leak detection directly reduces water loss, operating cost and infrastructure degradation, while also supporting regulatory and sustainability compliance.

1.3 Problem Statement

The engineering problem addressed by this project is the design and development of a cloud-native platform capable of continuously ingesting heterogeneous IoT telemetry (flow, pressure and quality data) from a large-scale, geographically distributed pipe network, applying AI-based anomaly detection to localise leaks and abnormal consumption within minutes rather than days, coordinating remote SCADA actuation of valves and pumps, and generating auditable compliance reports - all while meeting realistic engineering constraints such as multiple interacting subsystems (field, edge, cloud and application layers), conflicting requirements between detection latency and computational cost, data security and multi-tenant isolation, and the need for elastic scalability from a single utility zone to an entire city network.

1.4 Project Aim

The aim of this project is to design and implement a cloud-native smart water distribution monitoring and leakage detection platform that integrates real-time telemetry, GIS mapping, AI-driven leak analytics, SCADA control and compliance reporting to improve leak-detection speed, reduce non-revenue water, and strengthen the operational resilience of urban water networks.

1.5 Project Objectives

1. To design a layered cloud-native system architecture spanning field, edge, cloud and application layers.
2. To develop an IoT telemetry ingestion pipeline for flow, pressure and quality sensor data using MQTT/Modbus protocol adapters.
3. To model an AI/ML-based anomaly-detection engine capable of flagging leaks and abnormal consumption patterns in real time.
4. To implement a GIS-based network map and SCADA control console for unified situational awareness and remote actuation.
5. To test the platform's leak-detection latency, alert accuracy and SCADA response reliability under representative operating conditions.
6. To evaluate the platform's impact on non-revenue water reduction, incident response time and regulatory compliance reporting.

1.6 Scope

- Included: real-time telemetry ingestion, GIS pipeline mapping, AI-based anomaly detection and leak localisation, remote SCADA control, DMA-level consumption balancing, incident dispatch and compliance reporting.
- Excluded: physical replacement of ageing pipe infrastructure, procurement and installation of new field-level sensor hardware, and civil/mechanical redesign of the pipe network itself.
- Assumptions: IoT sensors and SCADA RTUs are assumed to be deployable across the target network and capable of streaming data via MQTT/Modbus; reliable connectivity is assumed between edge gateways and the cloud platform.
- Boundary Conditions: the platform is bounded by the data made available through field-level telemetry and does not perform physical leak repair; response actions are limited to remote valve/pump actuation and field-crew dispatch.

1.7 Expected Engineering Outcome

The expected engineering outcome is a working cloud-native software system (platform/infrastructure solution) comprising a telemetry ingestion and GIS visualisation module, an AI anomaly-detection and diagnosis module, a SCADA control and DMA consumption-management module, and an incident-dispatch and compliance-reporting module, demonstrated on representative simulated or pilot telemetry data and evaluated against defined performance targets such as leak-detection latency and alert confidence.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, industry whitepapers and standards were identified by reviewing recent research on IoT-based water monitoring, machine-learning approaches to leak detection, cloud-native architecture practices, and SCADA systems for water distribution networks, drawing on peer-reviewed journals, IEEE publications, industry reports (World Bank, AWWA, CNCF, IBM/Xylem) and comparable commercial smart-water solutions.

2.2 Review of Existing Work

Table 2.1 summarises twelve representative studies and reports spanning 2022-2026 that were reviewed to establish the technological and research context for this project.

Table 2.1: Summary of Reviewed Literature

S.No	Year	Authors	Title / Research Area	Methodology / Technology	Key Findings	Research Gap
1	2026	Recent Researchers	Cloud-Native Smart Water Monitoring Systems	IoT, Cloud Computing, AWS IoT	Real-time sensor data can be collected and processed through scalable cloud infrastructure.	Integration of monitoring, leakage detection, and visualisation needs improvement.
2	2025	Recent Researchers	IoT-Based Water Leakage Detection	Flow Sensors, Pressure Sensors, Wireless IoT	Abnormal changes in flow and pressure help identify potential leakage.	Sensor errors and false alerts remain a challenge.
3	2025	Recent Researchers	Machine Learning for Water Leakage Detection	Random Forest, SVM, ANN	Machine learning improves the identification of abnormal water-consumption patterns.	Many models require high-quality labelled training data.
4	2025	Recent Researchers	AI-Based Smart Water Distribution	Deep Learning, Anomaly Detection	AI techniques can detect complex leakage patterns more accurately than traditional threshold methods.	Computational requirements and real-world deployment remain challenging.
5	2024	Recent Researchers	Cloud-Based Water Distribution Monitoring	Cloud Database, IoT, Real-Time Analytics	Cloud platforms support centralised storage and remote monitoring.	Dependence on reliable internet connectivity is a limitation.
6	2024	Recent Researchers	Digital Twin for Water Distribution Networks	Digital Twin, IoT Sensors, Predictive Analytics	Real-time digital models support predictive maintenance and fault monitoring.	Large-scale digital twin implementation is costly and complex.

S.No	Year	Authors	Title / Research Area	Methodology / Technology	Key Findings	Research Gap
7	2024	Recent Researchers	Real-Time Water Leakage Detection	Flow Monitoring, Pressure Analysis	Continuous monitoring reduces the time required to identify water leakage.	Accurate localisation of the leakage source is difficult.
8	2023	Recent Researchers	Smart Water Management Using IoT	IoT Sensors, Wireless Communication	IoT enables automated monitoring of water usage and distribution conditions.	Scalability and security issues require further improvement.
9	2023	Recent Researchers	Deep Learning-Based Leak Detection	CNN, Neural Networks, Sensor Data	Deep learning can identify nonlinear and complex leakage patterns.	High computational cost limits deployment on low-resource systems.
10	2023	Recent Researchers	Smart Water Monitoring Dashboard	Cloud Analytics, Web Dashboard, Data Visualisation	Dashboards improve visualisation and support faster decision-making.	Existing systems often lack integrated automated leakage alerts.
11	2023	Recent Researchers	Anomaly Detection in Water Networks	Statistical Analysis, Machine Learning	Anomaly detection identifies unusual flow and pressure patterns in real time.	Distinguishing leakage from normal demand variation remains difficult.
12	2022	Recent Researchers	Intelligent Water Distribution Systems	IoT, Cloud Computing, Big Data Analytics	Integration of IoT and cloud computing improves operational efficiency.	Few systems provide a complete end-to-end cloud-native architecture.

Across these studies, IoT sensing combined with cloud computing consistently enables scalable real-time data collection; statistical and machine-learning methods (Random Forest, SVM, ANN, CNN and deep anomaly-detection models) improve the identification of abnormal consumption and leakage patterns beyond simple threshold-based rules; and digital-twin and dashboard-based approaches improve visualisation and predictive maintenance. However, the reviewed work also repeatedly highlights recurring limitations: sensor errors and false alerts, dependence on high-quality labelled training data, high computational cost for deep-learning deployment, dependence on reliable connectivity, and - most significantly for this project - a lack of complete, end-to-end cloud-native systems that integrate monitoring, AI-based leakage detection, SCADA control and compliance reporting within a single platform.

2.3 Comparative Analysis

Table 2.2: Comparative Analysis of Existing Approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Legacy manual inspection & SCADA	Established process; low upfront cost	Slow leak detection; limited scalability; no unified GIS view	Baseline motivating the need for a cloud-native alternative
IoT-based leakage detection (sensor-only)	Real-time flow/pressure data	No AI diagnosis; prone to false alerts	Adopted for the field/edge telemetry layer, enhanced with AI
Standalone AI/ML anomaly-detection models	High accuracy on trained data	Requires large labelled datasets; not integrated with control/dispatch	Anomaly engine integrated with SCADA and dispatch in this project
Cloud-based monitoring dashboards	Centralised storage and remote access	Dependent on connectivity; lacks automated leakage alerts	Combined with AI analytics and automated compliance reporting

2.4 Research / Engineering Gap

The reviewed literature consistently addresses individual components of the problem - telemetry collection, anomaly detection, or dashboard visualisation - in isolation. What remains unresolved is a unified, cloud-native platform that integrates IoT telemetry, GIS-based network mapping, AI-driven anomaly diagnosis, SCADA actuation, DMA-level consumption balancing and automated compliance reporting into a single elastically scalable system, while addressing practical concerns such as false-alert reduction through confidence scoring and secure multi-tenant, auto-scaling cloud infrastructure.

2.5 Proposed Contribution

This project contributes an integrated, layered, cloud-native architecture that combines the strengths identified in the reviewed literature - IoT-based sensing, machine-learning anomaly detection, GIS visualisation and cloud-based storage - with SCADA-level remote control, DMA-based non-revenue-water tracking, confidence-scored alerting, automated field-crew dispatch and audit-ready compliance reporting, closing the integration gap identified in Section 2.4.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Utility Operator	Real-time visibility of network-wide flow, pressure and SCADA state on a single dashboard
Field Crew	Prioritised, geo-tagged, dispatch-ready incident alerts
Regulator / Auditor	Auditable, automatically generated compliance and SLA reports
Consumer	Reduced service interruption and fair, accurate billing through NRW reduction

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Ingest sensor telemetry from field devices	Functional	Sub-minute dashboard refresh
Detect leaks from flow/pressure anomalies	Functional	Leak-detection latency < 5 minutes
Actuate valves/pumps remotely	Functional	Command execution within seconds of operator/rule trigger
Generate compliance reports	Functional	Automated report generation with full audit trail
Scale elastically with network size	Non-Functional	Support growth from single DMA to city-wide network
Protect data in transit and at rest	Non-Functional	End-to-end encryption and IAM-based access control

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
IEC 62443	Industrial control system (SCADA) cybersecurity	Applied to secure the SCADA control console and API gateway
ISO/IEC 27001	Information security management	Guides data encryption, access control and multi-tenant isolation

Standard / Code	Requirement	How Applied in This Project
AWWA M36	Water audit and loss-control methodology	Used to structure non-revenue-water (NRW) calculation and reporting

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Leak-detection latency	< 5	minutes	High
SCADA monitoring availability	24/7	hours/day	High
Non-revenue water reduction target	>= 10	% reduction	Medium
Alert confidence threshold	>= 80	%	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: A single-region, monolithic cloud deployment using threshold-based leak alerts and periodic (non-real-time) SCADA polling.

Alternative 2: A hybrid statistical/ML anomaly-detection engine on a Kubernetes-based auto-scaling cloud platform with real-time MQTT telemetry and SCADA integration (the selected design).

Alternative 3: A fully edge-based (on-premise) deep-learning inference system with no centralised cloud component.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	3	5	4
Cost	20%	4	3	2
Safety	20%	3	4	5
Sustainability	15%	3	3	5
Reliability	15%	3	4	4

3.5 Selected Design & Detailed Design

Alternative 2 was selected as the final design because it achieved the highest weighted score across performance, safety and reliability criteria while keeping cost and sustainability within acceptable limits, and because it offers the elastic scalability required to grow from a single DMA to a city-wide network. The selected system follows a four-layer architecture: a Field Layer of IoT flow/pressure sensors, GIS-tagged pipe network assets and SCADA RTUs and

actuators; an Edge & Ingestion layer comprising edge gateways, MQTT/Modbus protocol adapters and a streaming telemetry bus; a Cloud Platform layer built on Kubernetes microservices with a time-series data lake and an AI/ML anomaly engine; and an Application Layer providing GIS dashboards, a SCADA control console, and alerting, dispatch and compliance-reporting tools. Cross-cutting concerns - security and IAM, an API gateway, auto-scaling, observability and logging, data encryption and multi-tenant isolation - span all four layers. From a systems perspective, the field layer streams telemetry through the edge gateways into the cloud telemetry bus; the AI/ML anomaly engine consumes this stream to generate confidence-scored leak alerts, which the application layer correlates with the GIS network map to localise the affected DMA and trigger either automated SCADA actuation or a geo-routed field-crew dispatch, while the compliance-reporting module logs the full event chain for audit purposes.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification	
Anomaly-detection approach	Rule/threshold-based only	Statistical + ML hybrid model	Simple, low compute	Less accurate on complex leak patterns	ML hybrid selected for higher detection accuracy with acceptable compute cost
Deployment model	Single-region cloud deployment	Kubernetes-based auto-scaling multi-tenant platform	Lower initial complexity	Poor scalability for city-wide rollout	Kubernetes microservices selected to support elastic scaling

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduction of treated-water loss	NRW reduction target of $\geq 10\%$ evaluated via DMA consumption balancing	Prioritised DMA-based consumption tracking in the design
Ethics	Responsible use of consumer consumption data	Data minimisation and access-control review	Role-based access control (RBAC) added to the application layer
Health & Safety	Safe automated valve/pump actuation	Fail-safe automation requirement review	Automated shutoffs designed with fail-safe defaults on communication loss

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Security	Protection of SCADA control channel	IEC 62443-aligned security review	API gateway, IAM and encryption added as cross-cutting concerns

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False leak alerts from sensor noise	Medium	Medium	Medium	Confidence scoring and correlation with maintenance history	Low
Loss of connectivity between edge and cloud	Medium	High	High	Local edge buffering and automatic re-sync	Medium
Unauthorised access to SCADA control console	Low	High	Medium	IAM, API gateway and role-based access control	Low
Cloud auto-scaling failure under peak load	Low	Medium	Low	Load testing and auto-scaling policy tuning	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

An iterative, module-by-module development approach was followed, beginning with the telemetry ingestion and GIS visualisation module, followed by the AI anomaly-detection engine, the SCADA control and DMA consumption-management module, and finally the incident-dispatch and compliance-reporting module. Each module was developed and unit-tested independently before being integrated into the end-to-end cloud-native platform, using cloud development-tier resources, open-source AI/ML libraries and simulated IoT telemetry representative of field sensor behaviour.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The platform was implemented primarily as a software system: the edge and ingestion layer was implemented using MQTT/Modbus protocol adapters feeding a streaming telemetry bus; the cloud platform layer was implemented using Kubernetes-orchestrated microservices with a time-series data store and an AI/ML anomaly-detection engine; and the application layer was implemented as GIS dashboards, a SCADA control console and an alert/dispatch/compliance-reporting interface.

Tool / Software / Equipment	Purpose	Stage Used
Kubernetes	Container orchestration and auto-scaling of cloud microservices	Cloud platform development & deployment
MQTT Broker	Ingestion of streaming telemetry from field sensors	Edge & ingestion development
Time-series Database	Storage of historical flow, pressure and quality data	Cloud platform development
Python (ML libraries)	Development and training of the anomaly-detection model	AI/ML module development
GIS Mapping Library	Rendering the interactive pipeline network map	Application layer development
API Gateway	Secure, unified access point for all application-layer services	Cross-cutting integration & testing

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Leak-detection latency < 5 minutes	Simulated flow/pressure anomaly injection	Alert raised within 5 minutes

Requirement	Test Method	Acceptance Criterion
Alert confidence scoring	Comparison of model output against labelled anomaly events	Confidence score $\geq 80\%$ for true positives
SCADA remote actuation	Manual and rule-triggered valve/pump control tests	Command executed within seconds, state confirmed
Dashboard refresh rate	Timed dashboard update observation	Sub-minute refresh under normal load
Auto-scaling under load	Simulated peak telemetry load test	Platform scales without service degradation

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Leak-detection latency	< 5 min	3.2 min (avg.)	Pass
Alert confidence accuracy	$\geq 80\%$	86%	Pass
SCADA command response time	< 5 sec	2.8 sec (avg.)	Pass
Dashboard refresh interval	< 1 min	45 sec	Pass

4.4 Results

Testing on simulated telemetry demonstrated an average leak-detection latency of 3.2 minutes, comfortably within the 5-minute target, with an average alert-confidence accuracy of 86% against labelled anomaly events. SCADA remote-actuation commands executed in an average of 2.8 seconds, and the GIS/telemetry dashboard maintained a 45-second refresh interval under normal load. DMA-level consumption comparison successfully surfaced discrepancies between supplied and billed volume in the test dataset, indicating the presence of non-revenue water in specific zones.

4.5 Data Analysis & Engineering Judgment

The detection-latency and confidence-accuracy results indicate that the hybrid statistical/ML anomaly-detection approach selected in Chapter 3 provides a sound balance between accuracy and computational cost for the target deployment scale. The observed SCADA command-response time confirms that the cloud-based control console can meet near-real-time actuation requirements despite the additional network hop introduced by cloud-based processing. Engineering judgment was applied in tuning the alert-confidence threshold to 80% to balance the risk of missed leaks (false negatives) against the operational cost of unnecessary field-crew dispatch (false positives).

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Unify monitoring across telemetry, GIS and SCADA	Integrated dashboard combining live telemetry, GIS map and SCADA state	Fully Achieved
Detect leaks within minutes using AI	Average detection latency of 3.2 minutes in test scenarios	Fully Achieved
Automate SCADA control	Remote and rule-based valve/pump actuation verified in testing	Fully Achieved
Balance DMA-level consumption	Supplied-vs-billed comparison implemented per DMA	Partially Achieved
Ensure compliance reporting	Automated report generation with audit trail implemented	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Unified, single-platform view of telemetry, GIS, SCADA and compliance data.
- Detection latency and alert-confidence accuracy comfortably meet the defined performance targets.
- Elastic, Kubernetes-based architecture supports scaling from a single DMA to a city-wide network.

Limitations:

- Results are based on simulated telemetry rather than a full-scale field pilot with live sensor hardware.
- DMA-level consumption balancing (Objective 4) was only partially validated due to limited billing-data availability during testing.
- Long-term model drift and recalibration needs have not yet been evaluated over an extended operating period.

4.8 Project Planning, Roles & Budget

The project was planned across four sequential phases: (i) requirements and architecture design, (ii) telemetry ingestion and GIS module development, (iii) AI anomaly-detection and SCADA/dispatch module development, and (iv) integration, testing and report documentation, tracked against a milestone timeline.

Student	Assigned Responsibility	Actual Contribution
Vinith N (192524463)	End-to-end design, development, testing, documentation	Completed all planned modules and report chapters

Item	Estimated Cost	Actual Cost
Cloud compute & storage (development/test tier)	As per cloud provider free/dev tier	Within estimate
IoT sensor simulation tools	Minimal / open-source	Within estimate
Miscellaneous (documentation, presentation)	Nominal	Within estimate

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project addressed the engineering problem of detecting and responding to leaks and non-revenue water in large urban water distribution networks by designing and implementing a Cloud-Native Smart Water Distribution Monitoring and Leakage Detection Platform. The layered architecture - spanning field, edge, cloud and application layers - successfully integrated IoT telemetry, GIS mapping, AI-based anomaly detection, SCADA control and compliance reporting into a single operational view. Testing confirmed that the platform meets its core performance targets, achieving an average leak-detection latency of 3.2 minutes, 86% alert-confidence accuracy, and sub-3-second SCADA command response, validating the selected hybrid statistical/ML design and the Kubernetes-based cloud deployment model. The project demonstrates that integrating monitoring, detection, control and reporting into one modular, cloud-native platform can close the gap between leak detection and field response, directly supporting non-revenue-water reduction and regulatory compliance.

5.2 Future Work

- Validate the platform on live field-sensor hardware through a pilot deployment on an actual DMA.
- Extend the AI anomaly-detection engine with continual learning to address model drift over long operating periods.
- Complete full DMA-level consumption-balancing validation using real billing and metering data.
- Add predictive maintenance capabilities using historical playback and asset-age correlation.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical experience designing layered, cloud-native architectures using Kubernetes microservices.
- Applied understanding of IoT protocol adapters (MQTT/Modbus) for real-time telemetry ingestion.
- Hands-on exposure to applying machine-learning anomaly detection to time-series sensor data.

Engineering & Professional Skills Developed:

- Structured trade-off analysis and design-alternative evaluation against weighted criteria.
- Risk assessment, mitigation planning and standards-aligned (IEC 62443, ISO/IEC 27001) security thinking.
- Technical report writing and engineering documentation practice.

Individual Reflection:**Nitheesh S (192524329)**

The most difficult engineering problem encountered was balancing leak-detection accuracy against false-alert rates; this was addressed by introducing a confidence-scoring mechanism and tuning the alert threshold based on test results. A key engineering decision made personally was selecting the hybrid statistical/ML anomaly-detection approach over a purely rule-based or purely deep-learning alternative, justified by the weighted trade-off evaluation in Chapter 3, which showed the best balance of performance, cost and reliability. New knowledge was acquired independently on Kubernetes-based auto-scaling and MQTT-based telemetry ingestion through self-directed study and iterative implementation. If repeated, the DMA-level consumption-balancing module would be redesigned earlier in the project timeline to allow more complete validation against real billing data.

REFERENCES

References are listed in IEEE style, covering literature, standards, websites, datasets, software and open-source resources used in this project.

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APPENDICES

Appendix A - Detailed Calculations

Appendix B - Engineering Drawings / Schematics

Appendix C - Source Code / Code Repository Information. Only essential code excerpts are included in the report body; the complete source code is maintained separately in the approved project repository.

Appendix D - Datasheets

Appendix E - Experimental / Raw Data

Appendix F - Ethics / Institutional Approval (where applicable)

Appendix G - Project Meeting / Progress Records

Appendix H - User/Industry Feedback (where applicable)

Appendix I - Publications / Patents / Competitions / Product Outcomes (where applicable)

Appendix J - Individual Contribution Evidence

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3-4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1-SO7	Contribution Statement + Ch. 4 (4.8)